Factorial ANOVA Demo

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Factorial ANOVA | Research Context Prompt

A local dating expert was interested in studying first-date satisfaction and whether it depended on variables like (1) date attire (i.e., Athletic Wear, Leisure Wear, or Dressed Up) and (2) dating application used (OKCupid, Tinder).

Participants consented to be in their study after setting up their first date using either OKCupid or Tinder (this variable could not be randomly assigned). Upon meeting inclusion criteria and joining the study, the dating expert randomly assigned each participant to one of the three attire conditions.

The research question was: Does first-date satisfaction differ interactively when considering dating attire and the mobile application used?

3 x 2 Factorial ANOVA | Data

```
library(psych)
library(tidyverse)
library(jmv)
library(ggpubr)
library(ggrain)
library(apaTables)
library(ez)
library(rstatix)

dat_date <- read.csv("FACTORIAL_DEMO.csv")

# Set IVs to factors using as.factor() or factor()
dat_date$Application<-as.factor(dat_date$Application)
dat_date$Outfit<-as.factor(dat_date$Outfit)

# Assess number of participants and variables in data set
dim(dat_date)
```

```
## [1] 120   3
```

```
head(dat_date,4)
```

```
##   Application       Outfit Satisfaction
## 1     OkCupid AthleticWear            6
## 2     OkCupid AthleticWear            6
## 3     OkCupid AthleticWear            7
## 4     OkCupid AthleticWear            8
```

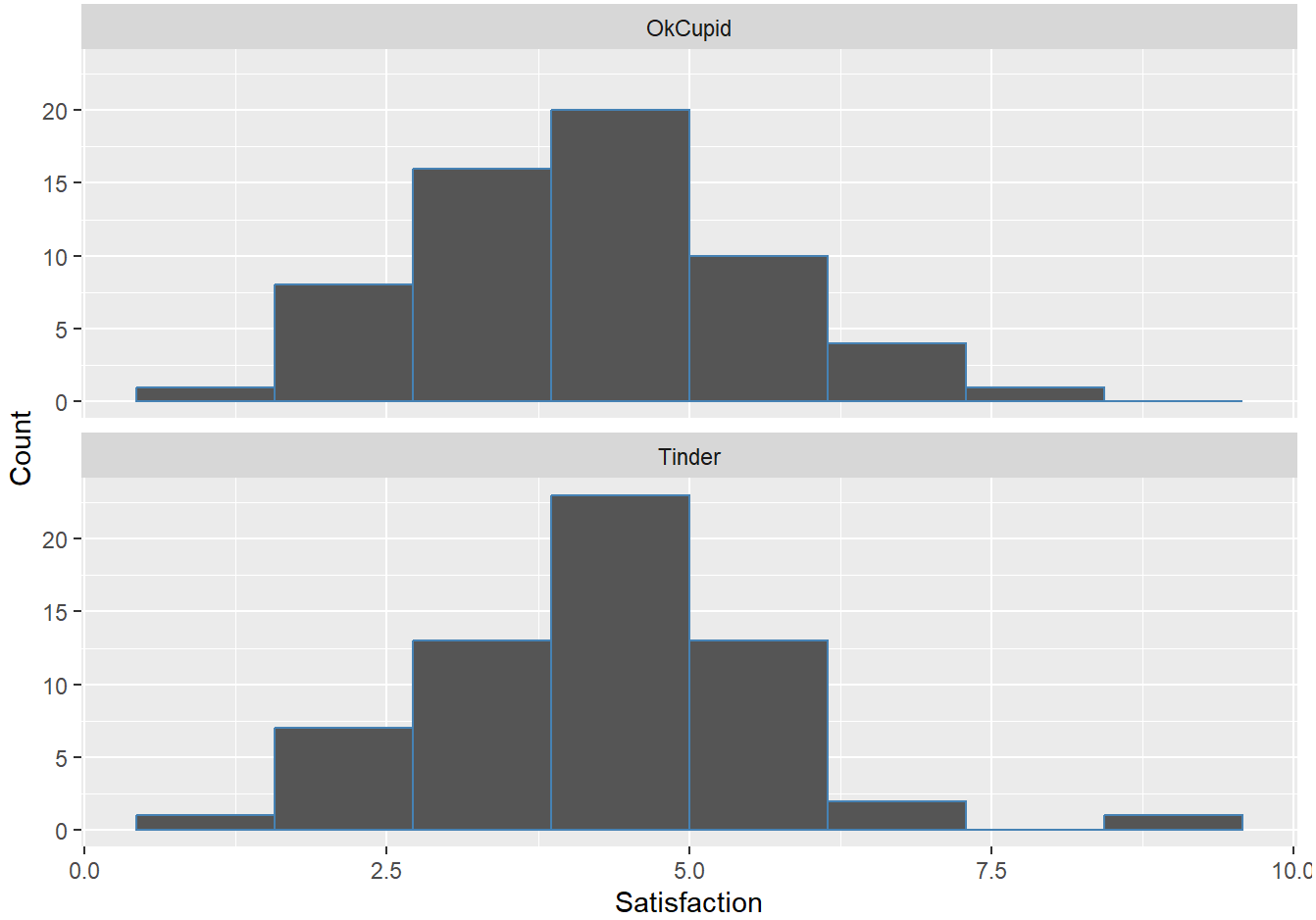
Factorial ANOVA | Descriptive Stats

```
# Descriptive Stats
describeBy(dat_date$Satisfaction,
           list(dat_date$Outfit,dat_date$Application),
           mat=TRUE)
```

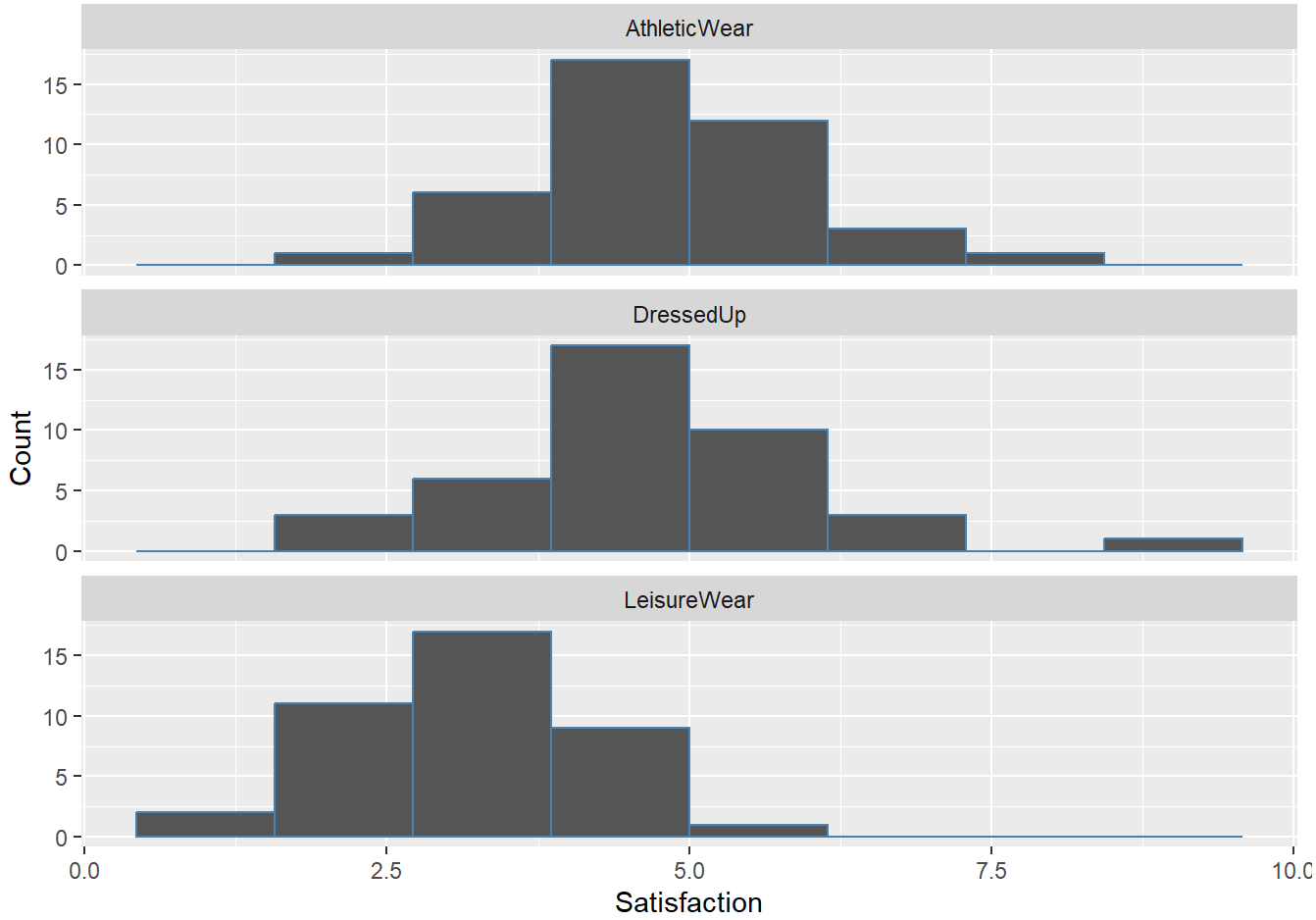
```
##     item       group1  group2 vars  n mean        sd median trimmed    mad min
## X11    1 AthleticWear OkCupid    1 20 5.80 1.0563094      6  5.8125 0.7413   4
## X12    2    DressedUp OkCupid    1 20 3.75 1.2513151      4  3.6875 1.4826   2
## X13    3  LeisureWear OkCupid    1 20 2.90 0.9119095      3  2.8750 0.7413   1
## X14    4 AthleticWear  Tinder    1 20 3.90 1.0208356      4  3.8125 1.4826   2
## X15    5    DressedUp  Tinder    1 20 5.80 1.1050125      6  5.7500 0.7413   4
## X16    6  LeisureWear  Tinder    1 20 3.05 1.1909748      3  2.9375 1.4826   1
##     max range       skew    kurtosis        se
## X11   8     4 -0.1323583 -0.49459060 0.2361980
## X12   7     5  0.6028951  0.08746646 0.2798026
## X13   5     4  0.1819801 -0.09670045 0.2039092
## X14   6     4  0.4681251 -0.29305249 0.2282658
## X15   9     5  0.8182151  1.42879608 0.2470883
## X16   6     5  0.6202259 -0.05741963 0.2663101
```

Factorial ANOVA | Histogram

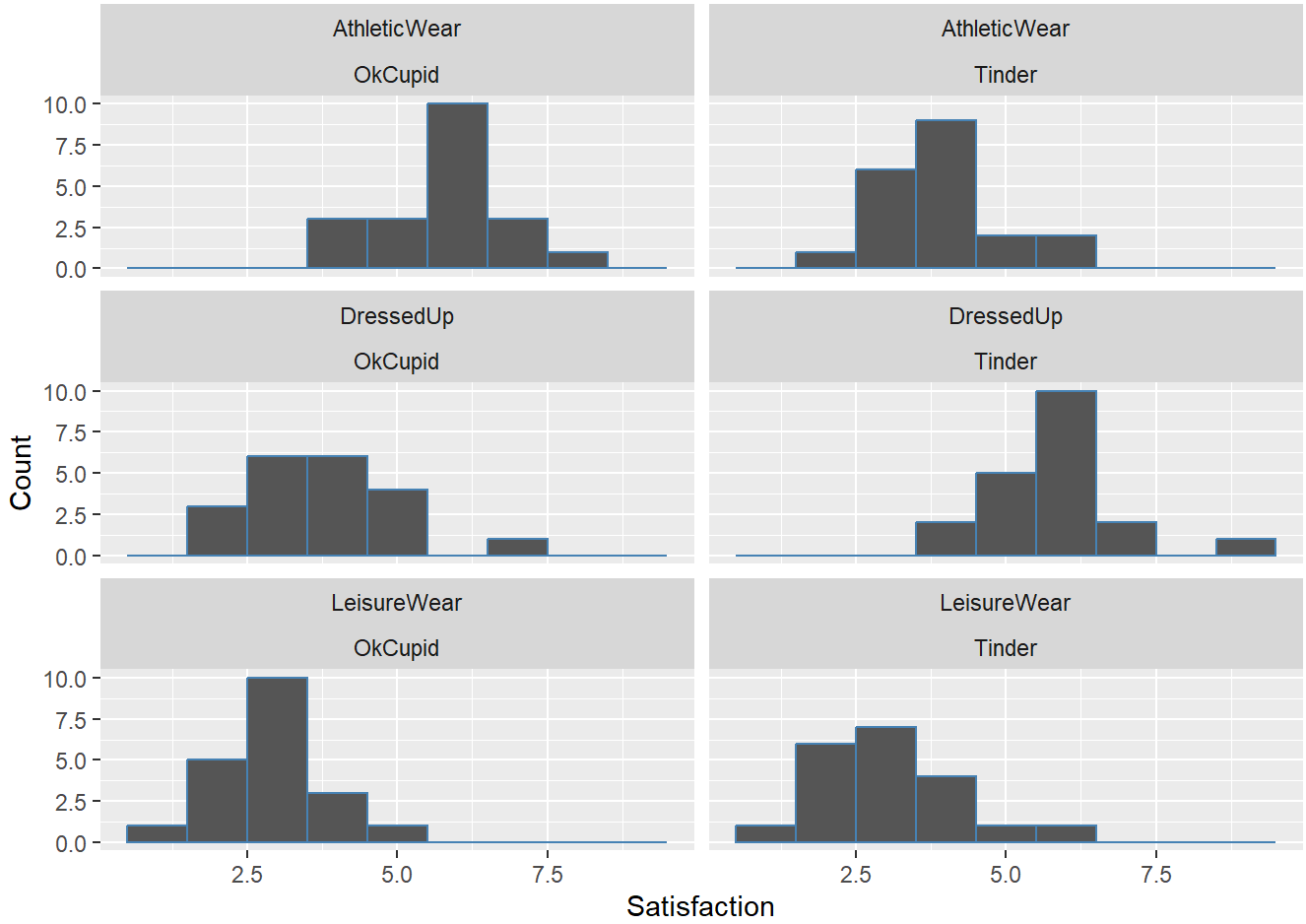
```
ggplot(data = dat_date,
       mapping = aes(x = Satisfaction)) +
  geom_histogram(bins = 8, color = "steelblue") +
  labs(y = "Count", x = "Satisfaction")+
  facet_wrap(dat_date$Application,ncol=1)
```




```
ggplot(data = dat_date,
       mapping = aes(x = Satisfaction)) +
  geom_histogram(bins = 8, color = "steelblue") +
  labs(y = "Count", x = "Satisfaction")+
  facet_wrap(dat_date$Outfit,ncol=1)
```

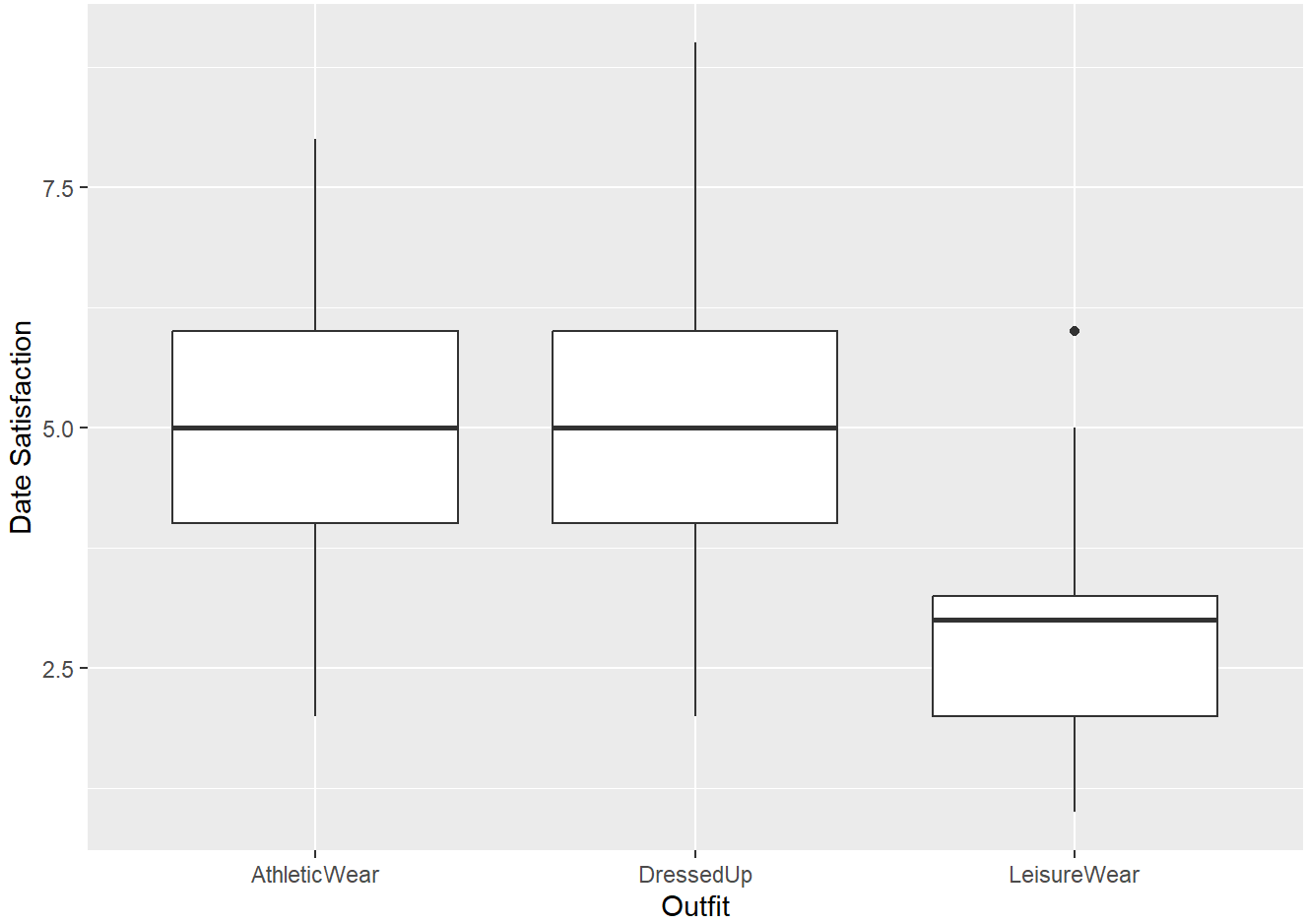



```
ggplot(data = dat_date,
       mapping = aes(x = Satisfaction)) +
  geom_histogram(bins = 9, color = "steelblue") +
  labs(y = "Count",
       x = "Satisfaction")+
  facet_wrap(~dat_date$Outfit+dat_date$Application,ncol = 2)
```

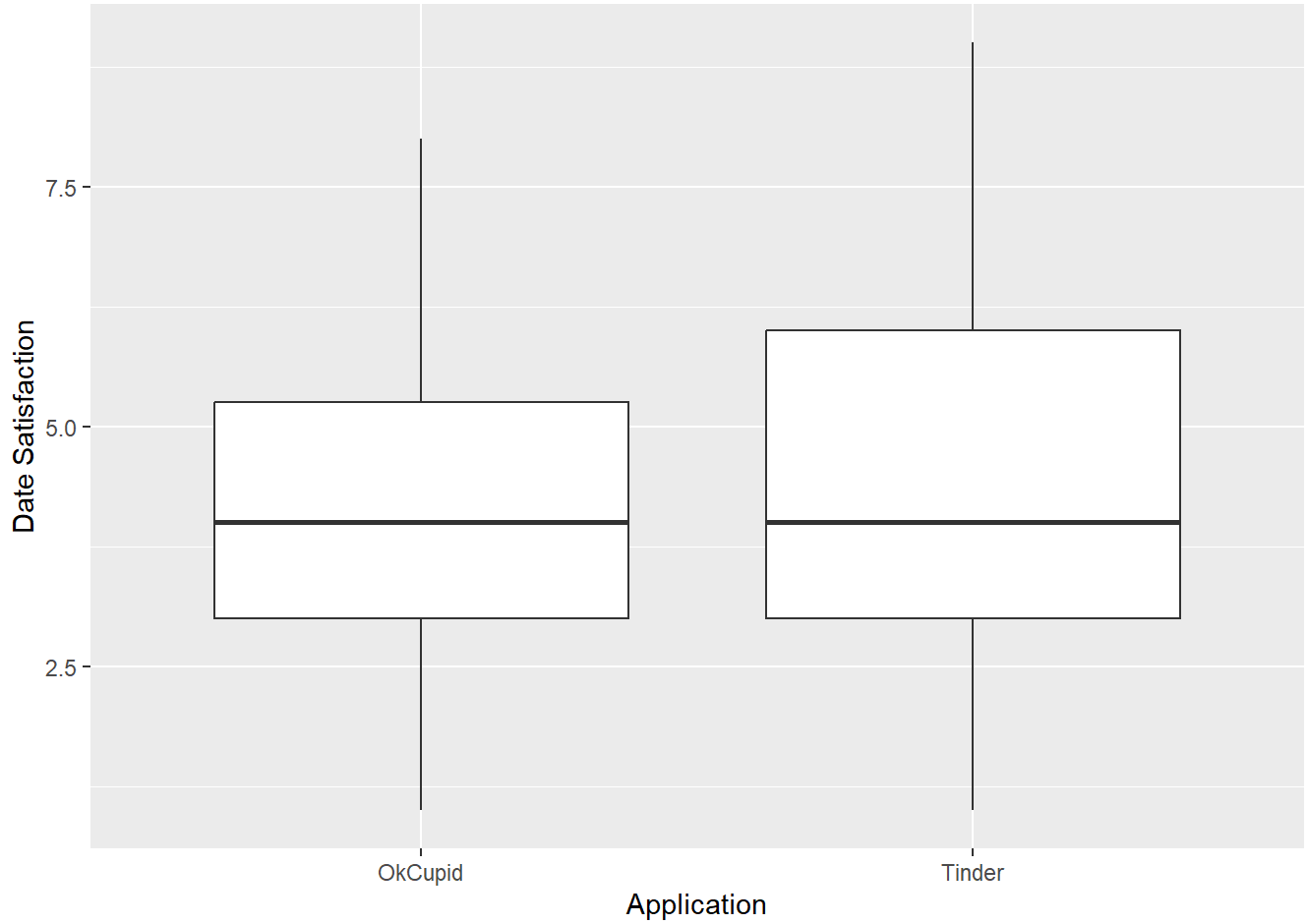


Factorial ANOVA | Boxplots and Raincloud Visualizations

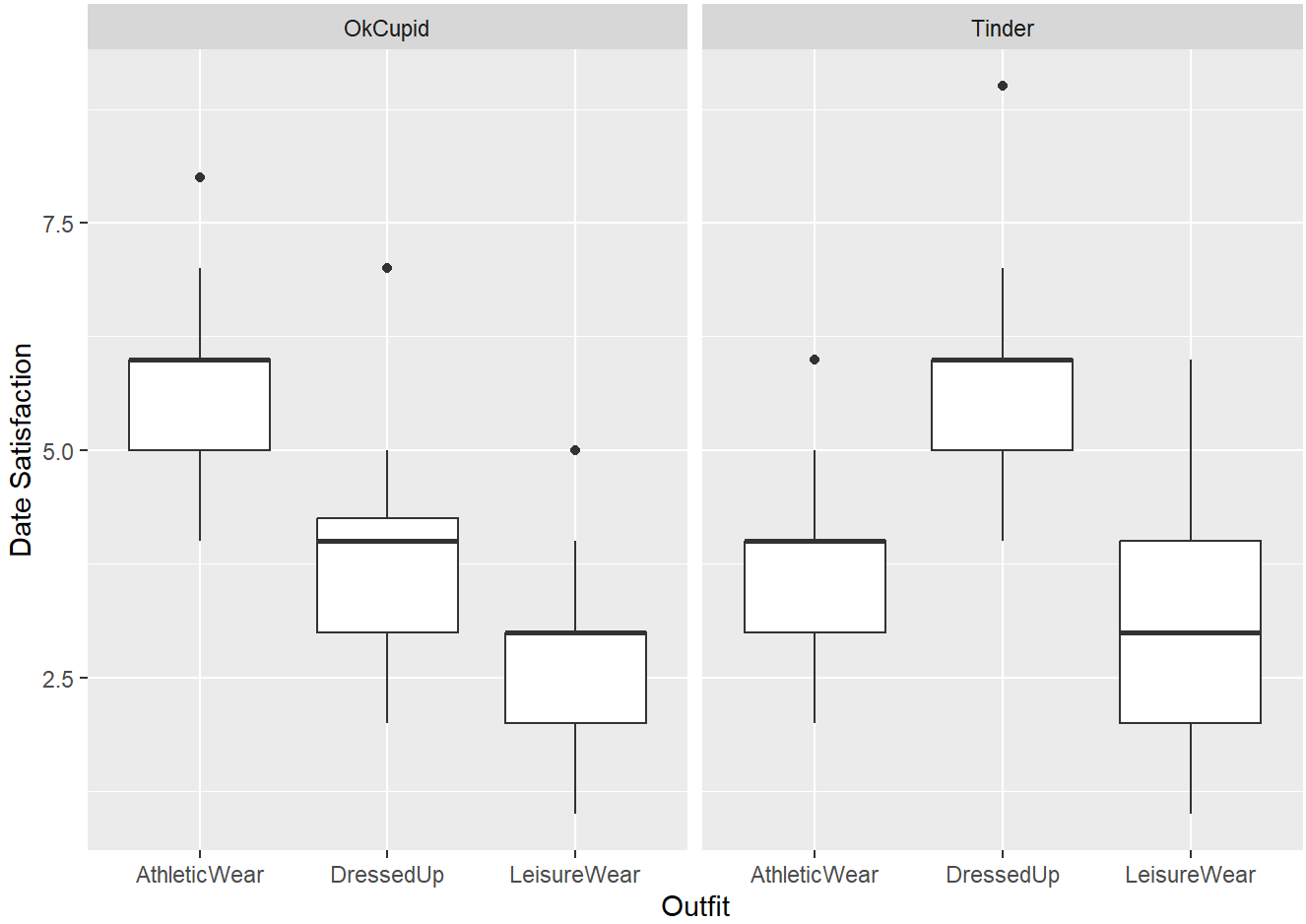
```
ggplot(data = dat_date,
       mapping = aes(y = Satisfaction, x = Outfit)) +
  geom_boxplot() +
    labs(y = "Date Satisfaction")
```



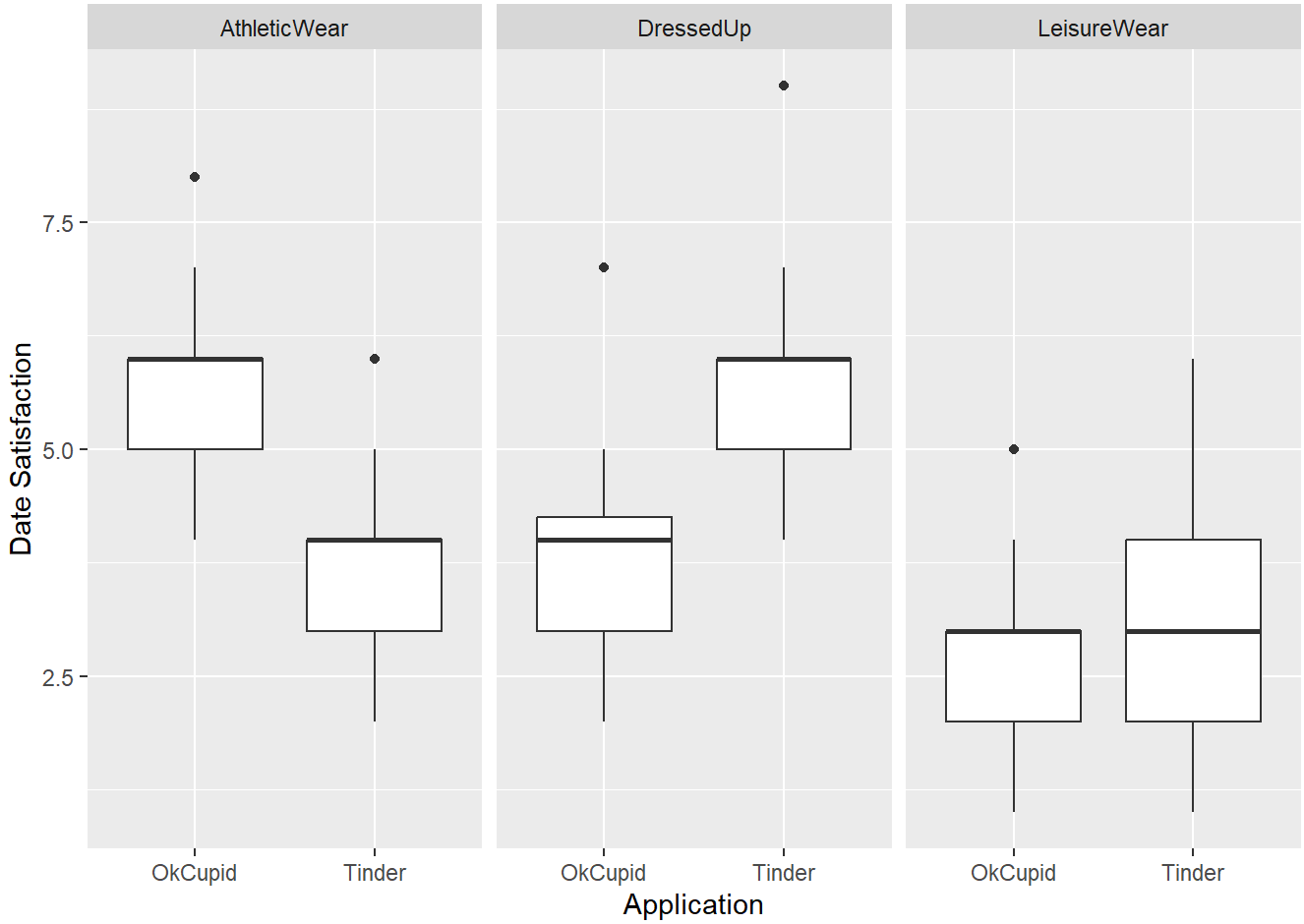

```
ggplot(data = dat_date,
       mapping = aes(y = Satisfaction, x = Application)) +
  geom_boxplot() +
    labs(y = "Date Satisfaction")
```



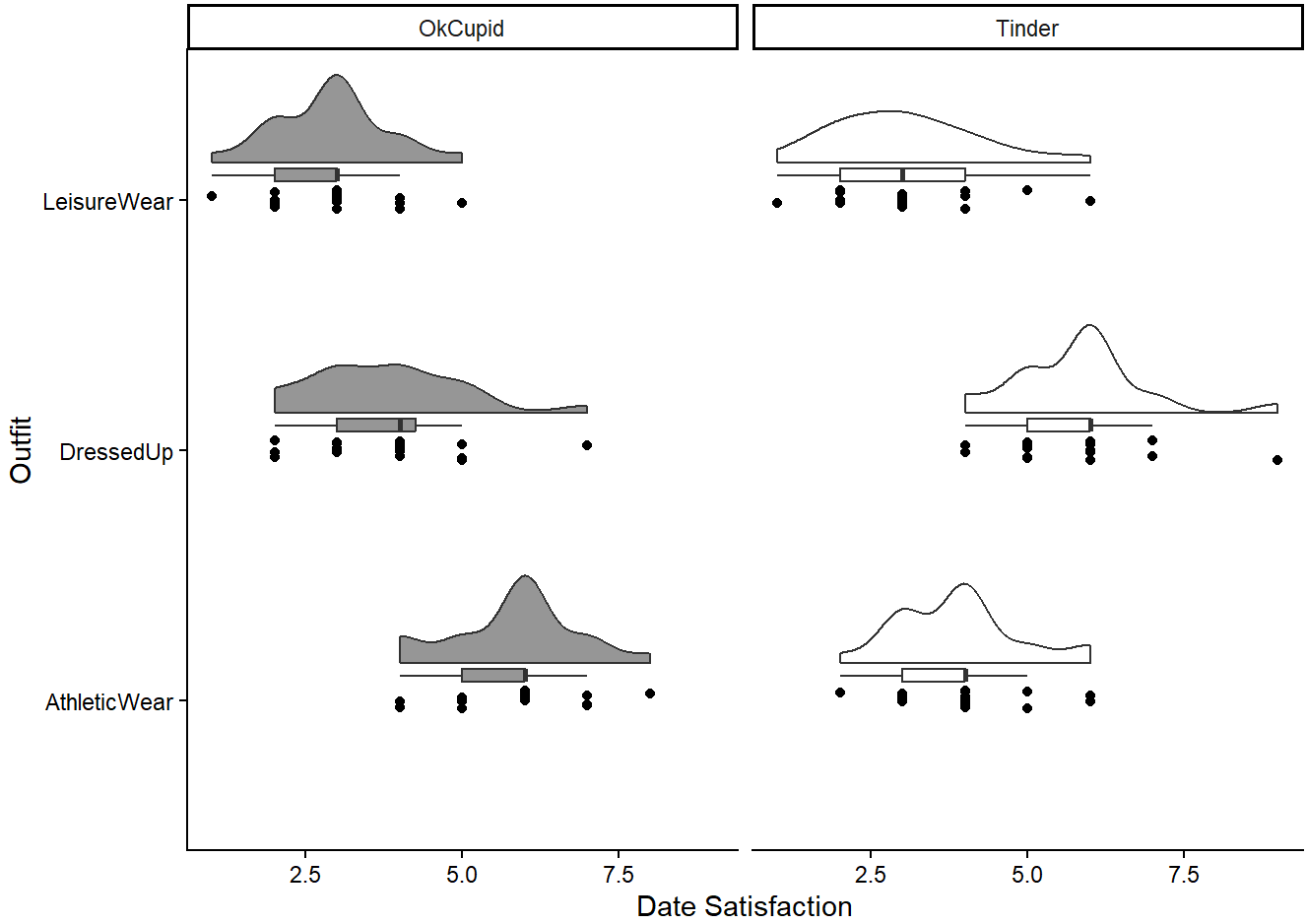

```
ggplot(data = dat_date,
       mapping = aes(y = Satisfaction, x = Outfit)) +
  geom_boxplot() +
  facet_wrap(~Application)+
    labs(y = "Date Satisfaction")
```




```
ggplot(data = dat_date,
       mapping = aes(y = Satisfaction, x = Application)) +
  geom_boxplot() +
  facet_wrap(~Outfit) +
    labs(y = "Date Satisfaction")
```




```
ggplot(data = dat_date,
       aes(x = Outfit, y = Satisfaction,
           fill = Application))+
  geom_rain(show.legend=FALSE) +
  facet_wrap(~Application)+
  scale_colour_grey(aesthetics = "fill",
                    start = .6,
                    end = 1) +
  coord_flip() +
  ylab("Date Satisfaction")+
  xlab("Outfit")+
  theme_classic()
```

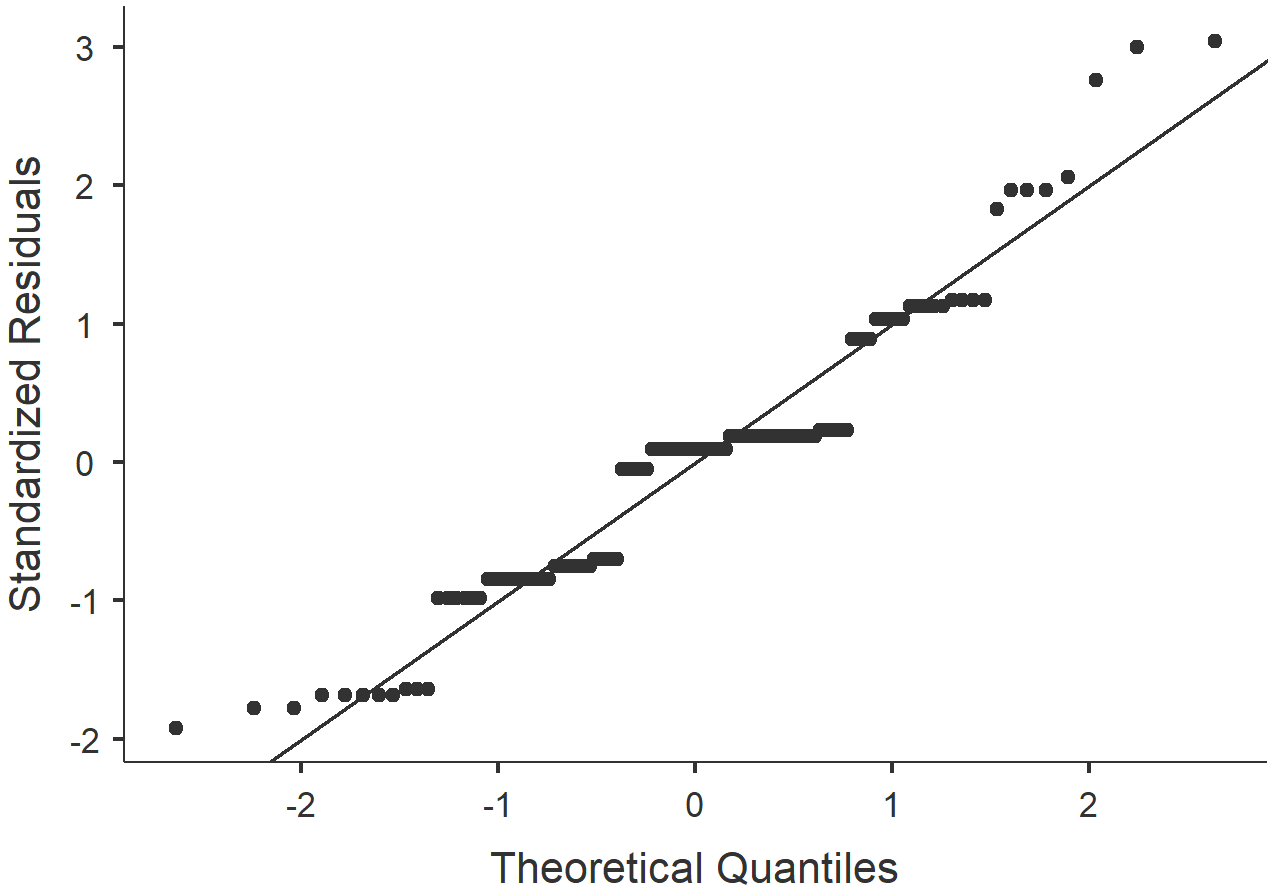


Factorial ANOVA | ANOVA() Omnibus Assumption Checks

```
# Omnibus Assumption Checking
ANOVA(data = dat_date,
      dep = 'Satisfaction',
      factors = c('Application','Outfit'),
      homo = TRUE,
      norm = TRUE,
      qq = TRUE)
```

```
##
##  ANOVA
##
##  ANOVA - Satisfaction
##  ─────────────────────────────────────────────────────────────────────────────────────────
##                           Sum of Squares    df     Mean Square    F             p
##  ─────────────────────────────────────────────────────────────────────────────────────────
##    Application                0.3000000      1      0.3000000     0.2501829     0.6179105
##    Outfit                    90.1500000      2     45.0750000    37.5899781    < .0000001
##    Application:Outfit        78.0500000      2     39.0250000    32.5446233    < .0000001
##    Residuals                136.7000000    114      1.1991228
##  ─────────────────────────────────────────────────────────────────────────────────────────
##
##
##  ASSUMPTION CHECKS
##
##  Homogeneity of Variances Test (Levene's)
##  ────────────────────────────────────────────
##    F            df1    df2    p
##  ────────────────────────────────────────────
##    0.5146008      5    114    0.7647589
##  ────────────────────────────────────────────
##
##
##  Normality Test (Shapiro-Wilk)
##  ──────────────────────────────
##    Statistic    p
##  ──────────────────────────────
##    0.9379552    0.0000312
##  ──────────────────────────────
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## ℹ Please use the `linewidth` argument instead.
## ℹ The deprecated feature was likely used in the jmvcore package.
##   Please report the issue at <https://github.com/jamovi/jmvcore/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



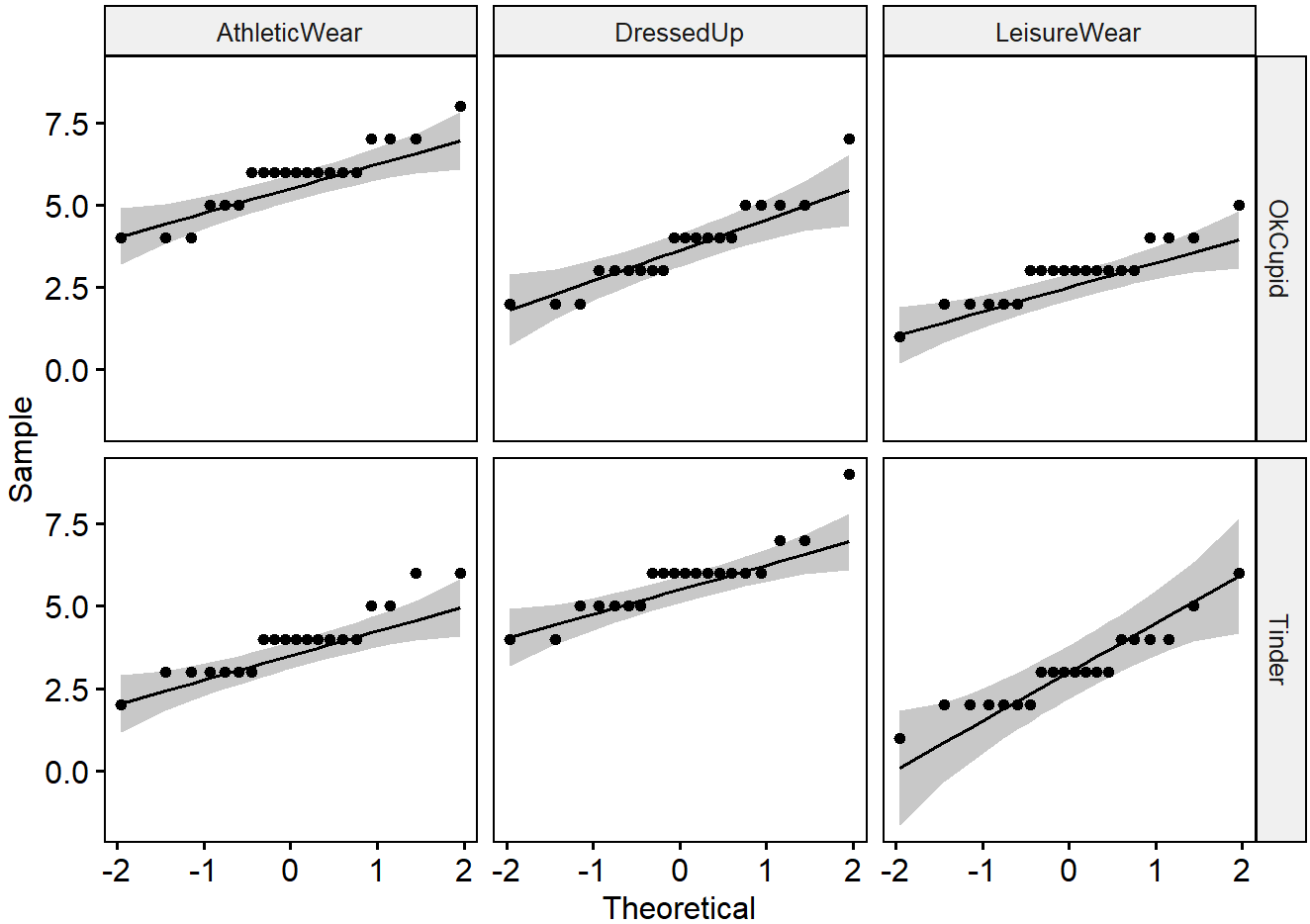
Factorial ANOVA | Group Level Assumption Checks

```
dat_date %>%
  group_by(Application,Outfit) %>%
  shapiro_test(Satisfaction)
```

```
## # A tibble: 6 × 5
##   Application Outfit       variable     statistic       p
##   <fct>       <fct>        <chr>            <dbl>   <dbl>
## 1 OkCupid     AthleticWear Satisfaction     0.890 0.0273
## 2 OkCupid     DressedUp    Satisfaction     0.909 0.0623
## 3 OkCupid     LeisureWear  Satisfaction     0.898 0.0381
## 4 Tinder      AthleticWear Satisfaction     0.887 0.0235
## 5 Tinder      DressedUp    Satisfaction     0.856 0.00663
## 6 Tinder      LeisureWear  Satisfaction     0.915 0.0796
```

Factorial ANOVA | Group Level Assumption Checks

```
ggqqplot(dat_date,
         'Satisfaction',
         facet.by = c("Application","Outfit"))
```



Factorial ANOVA | Conducting the ANOVA

```
options(digits = 3)
ANOVA(data = dat_date,
      dep = 'Satisfaction',
      factors = list('Application','Outfit'),
      effectSize = 'partEta',
      postHoc = ~Outfit + Outfit:Application,
      postHocCorr = 'bonf',
      postHocES = 'd',
      postHocEsCi = TRUE,
      emMeans = ~Application + Outfit + Outfit:Application,
      emmPlots = FALSE,
      emmPlotData = TRUE,
      emmTables = TRUE)
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
##
##  ANOVA
##
##  ANOVA - Satisfaction
##  ─────────────────────────────────────────────────────────────────────────────────────────────
##                          Sum of Squares    df     Mean Square    F         p         η²p
##  ─────────────────────────────────────────────────────────────────────────────────────────────
##    Application                    0.300      1          0.300     0.250     0.618    0.002
##    Outfit                        90.150      2         45.075    37.590    < .001    0.397
##    Application:Outfit            78.050      2         39.025    32.545    < .001    0.363
##    Residuals                    136.700    114          1.199
##  ─────────────────────────────────────────────────────────────────────────────────────────────
##
##
##  POST HOC TESTS
##
##  Post Hoc Comparisons - Outfit
##  ─────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────
##    Outfit               Outfit         Mean Difference    SE       df     t        p-bonferro
ni    Cohen's d    Lower     Upper
##  ─────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────
##    AthleticWear    -    DressedUp               0.0750    0.245    114    0.306           1.0
00       0.0685    -0.375    0.512
##                    -    LeisureWear             1.8750    0.245    114    7.657    < .001
1.7123     1.216    2.209
##    DressedUp       -    LeisureWear             1.8000    0.245    114    7.351    < .001
1.6438     1.151    2.136
##  ─────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────
##    Note. Comparisons are based on estimated marginal means
##
##
##  Post Hoc Comparisons - Outfit:Application
##  ─────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────────────────────────────────────────────────────────
##    Outfit          Application         Outfit          Application    Mean Difference    SE
df     t            p-bonferroni    Cohen's d    Lower       Upper
##  ─────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────────────────────────────────────────────────────────
##    AthleticWear    OkCupid        -    AthleticWear    Tinder                  1.900    0.34
6    114        5.487    < .001        1.735     1.06857     2.402
##                                   -    DressedUp       OkCupid                 2.050    0.34
6    114        5.920    < .001        1.872     1.19920     2.545
##                                   -    DressedUp       Tinder               -1.94e-15    0.34
6    114    -5.61e-15           1.000    -8.88e-16    -0.62644     0.626
##                                   -    LeisureWear     OkCupid                 2.900    0.34
6    114        8.375    < .001        2.648     1.93195     3.365
##                                   -    LeisureWear     Tinder                  2.750    0.34
6    114        7.941    < .001        2.511     1.80351     3.219
##                     Tinder         -    DressedUp       OkCupid                 0.150    0.34
```

```
6    114        0.433           1.000       -0.137    -0.76368     0.490
##                                     -    DressedUp       Tinder                  -1.900    0.34
6    114       -5.487    < .001       -1.735    -2.40161    -1.069
##                                     -    LeisureWear     OkCupid                  1.000    0.34
6    114        2.888           0.070       -0.913    -1.55100    -0.275
##                                     -    LeisureWear     Tinder                   0.850    0.34
6    114        2.455           0.234        0.776     0.14156     1.411
##    DressedUp       OkCupid        -    DressedUp       Tinder                  -2.050    0.34
6    114       -5.920    < .001       -1.872    -2.54494    -1.199
##                                     -    LeisureWear     OkCupid                  0.850    0.34
6    114        2.455           0.234        0.776     0.14156     1.411
##                                     -    LeisureWear     Tinder                   0.700    0.34
6    114        2.021           0.684        0.639     0.00721     1.271
##                     Tinder         -    LeisureWear     OkCupid                  2.900    0.34
6    114        8.375    < .001       -2.648    -3.36464    -1.932
##                                     -    LeisureWear     Tinder                   2.750    0.34
6    114        7.941    < .001        2.511     1.80351     3.219
##    LeisureWear     OkCupid        -    LeisureWear     Tinder                  -0.150    0.34
6    114       -0.433           1.000       -0.137    -0.76368     0.490
## ──────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────────────────────────────────
##    Note. Comparisons are based on estimated marginal means
##
##
##  ESTIMATED MARGINAL MEANS
##
##  APPLICATION
##
##  Estimated Marginal Means - Application
## ──────────────────────────────────────────────────
##    Application    Mean    SE       Lower    Upper
## ──────────────────────────────────────────────────
##    OkCupid        4.15    0.141     3.87     4.43
##    Tinder         4.25    0.141     3.97     4.53
## ──────────────────────────────────────────────────
##
##
##  OUTFIT
##
##  Estimated Marginal Means - Outfit
## ────────────────────────────────────────────────────
##    Outfit          Mean    SE       Lower    Upper
## ────────────────────────────────────────────────────
##    AthleticWear    4.85    0.173     4.51     5.19
##    DressedUp       4.78    0.173     4.43     5.12
##    LeisureWear     2.98    0.173     2.63     3.32
## ────────────────────────────────────────────────────
##
##
##  OUTFIT:APPLICATION
##
##  Estimated Marginal Means - Outfit:Application
```

```
## ────────────────────────────────────────────────────────────────
##   Application    Outfit          Mean    SE       Lower    Upper
## ────────────────────────────────────────────────────────────────
##   OkCupid        AthleticWear    5.80    0.245     5.31     6.29
##                  DressedUp       3.75    0.245     3.26     4.24
##                  LeisureWear     2.90    0.245     2.41     3.39
##   Tinder         AthleticWear    3.90    0.245     3.41     4.39
##                  DressedUp       5.80    0.245     5.31     6.29
##                  LeisureWear     3.05    0.245     2.56     3.54
## ────────────────────────────────────────────────────────────────
```

Factorial ANOVA | ANOVA Output: Interaction Simple Effects Code

```
model <- lm(Satisfaction~Application*Outfit, dat_date)
# Effect of Outfit across each Application
# When statistically significant interaction occurs we assess simple effects
one.way <- dat_date %>%
  group_by(Outfit) %>%
  anova_test(dv = Satisfaction,
             between = Application,
             error = model)%>%
  get_anova_table()
one.way
```

```
## # A tibble: 3 × 8
##   Outfit       Effect        DFn   DFd      F            p `p<.05`   ges
## * <fct>        <chr>       <dbl> <dbl>  <dbl>        <dbl> <chr>   <dbl>
## 1 AthleticWear Application     1   114 30.1   0.00000025   "*"     0.209
## 2 DressedUp    Application     1   114 35.0   0.0000000346 "*"     0.235
## 3 LeisureWear  Application     1   114  0.188 0.666        ""      0.002
```

```
one.way2 <- dat_date %>%
  group_by(Application) %>%
  anova_test(dv = Satisfaction,
             between = Outfit,
             error = model)%>%
  get_anova_table()
one.way2
```

```
## # A tibble: 2 × 8
##   Application Effect   DFn   DFd     F        p `p<.05`   ges
## * <fct>       <chr>  <dbl> <dbl> <dbl>    <dbl> <chr>   <dbl>
## 1 OkCupid     Outfit     2   114  37.1 3.97e-13 *       0.394
## 2 Tinder      Outfit     2   114  33.1 4.73e-12 *       0.367
```

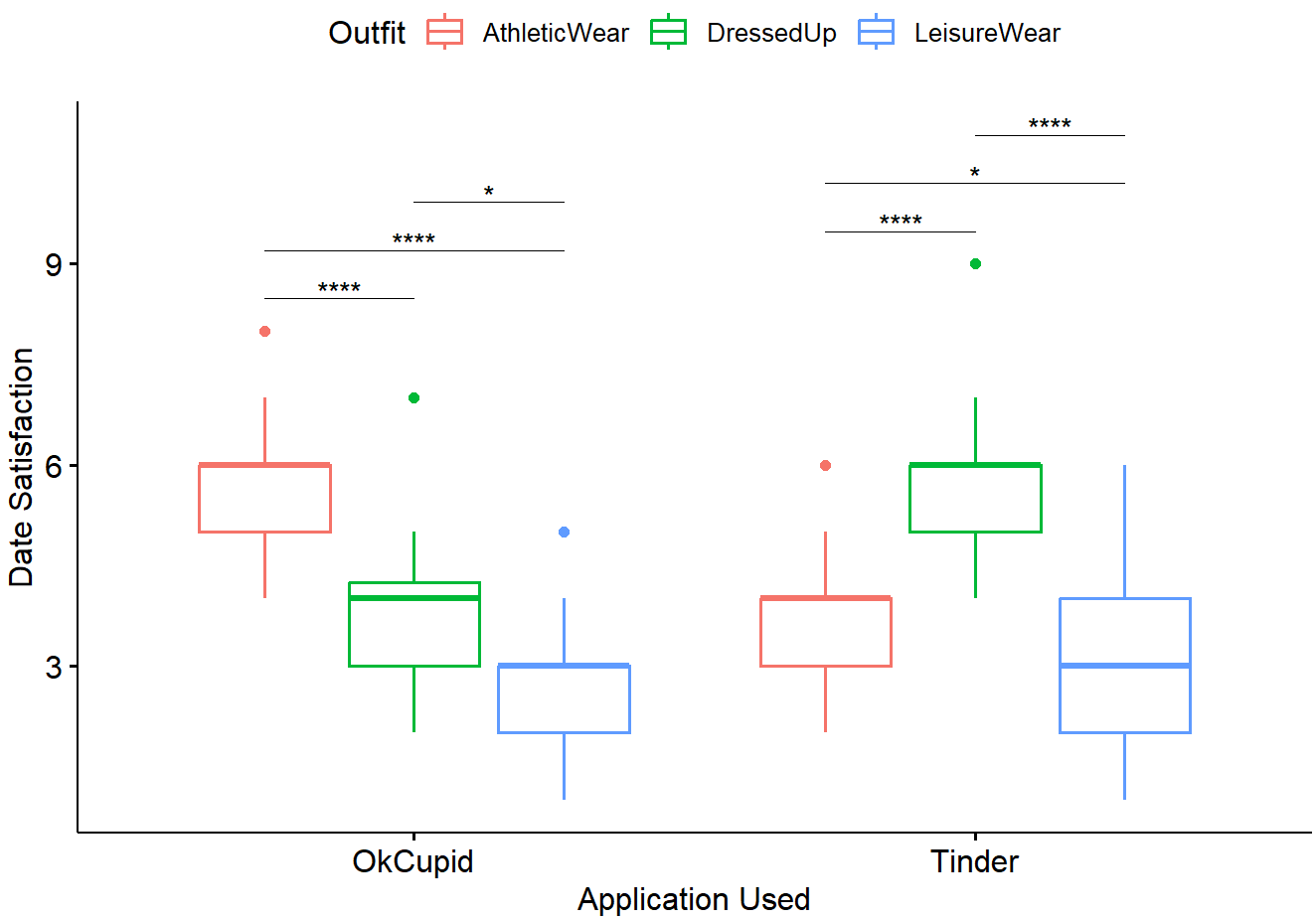
The simple main effect of Outfit on date satisfaction was statistically significant for both OKCupid and Tinder users ($p < .001$). In other words, there was a statistically significant difference in mean date satisfaction scores between OKCupid users based on date attire [$F(2,114) = 37.07$, $p < .001$]. And the same conclusion holds true for Tinder users [$F(2,114) = 33.07$, $p < .001$]. We would need to do further post-hoc testing to figure out where the differences occur.

BUT this is not as informative as looking at the simple main effects of Application on date satisfaction for Athletic, Dressed Up, and Leisure Wear conditions. There was no statistically significant difference between applications when participants wore leisure wear [$F(1,114) = 0.19$, $p > .05$]. But significant differences arose across applications for those wearing Athletic Wear [$F(1,114) = 30.11$, $p < .001$, $\eta_p^2 = .21$) and those that Dressed Up [$F(1,114) = 35.05$, $p < .001$, $\eta_p^2 = .24$].

Factorial ANOVA | Professional ANOVA Visualization Code

```
pwc1 <- dat_date %>%
  group_by(Application) %>%
  emmeans_test(Satisfaction~Outfit,
               p.adjust.method = "bonferroni") %>%
  add_xy_position(x = "Application")

ggboxplot(dat_date,
          x = "Application",
          y = "Satisfaction",
          color = "Outfit") +
  stat_pvalue_manual(pwc1,
                     hide.ns = FALSE,
                     tip.length = 0) +
  labs(y = "Date Satisfaction",
       x = "Application Used")
```



```
pwc2 <- dat_date %>%
  group_by(Outfit) %>%
  emmeans_test(Satisfaction~Application,
               p.adjust.method = "bonferroni") %>%
  add_xy_position(x = "Outfit")

ggboxplot(dat_date,
          x = "Outfit",
          y = "Satisfaction",
          color = "Application") +
  stat_pvalue_manual(pwc2,
                     hide.ns = FALSE,
                     tip.length = 0) +
  labs(
    caption = get_pwc_label(pwc2),
    y = "Date Satisfaction",
    x = "Outfit Worn")
```

Factorial ANOVA | APA Style ANOVA Tables

```
# ANOVA Source Table
aovm3<-ANOVA(data = dat_date,
             dep = 'Satisfaction',
             factors = list('Application','Outfit'),
             effectSize = 'partEta')$model

apa.aov.table(aovm3,
              table.number=5,
              conf.level=.95)
```

```
##
##
## Table 5
##
## ANOVA results using Satisfaction as the dependent variable
##
##
##             Predictor      SS  df      MS       F    p partial_eta2
##           (Intercept) 2116.80   1 2116.80 1765.29 .000
##           Application    0.30   1    0.30    0.25 .618          .00
##                Outfit   90.15   2   45.08   37.59 .000          .40
##  Application x Outfit   78.05   2   39.02   32.54 .000          .36
##                 Error  136.70 114    1.20
##  CI_95_partial_eta2
##
##          [.00, .05]
##          [.25, .50]
##          [.22, .47]
##
##
## Note: Values in square brackets indicate the bounds of the 95% confidence interval for partia
l eta-squared
```

```
#Generate Marginal Means Table
apa.2way.table(Outfit,
               Application,
               Satisfaction,
               dat_date,
               table.number=6,
               show.conf.interval = TRUE)
```

```
##
##
## Table 6
##
## Means and standard deviations for Satisfaction as a function of a 3(Outfit) X 2(Application)
design
##
##                          M     M_95%_CI   SD
##  Application:OkCupid
##               Outfit
##         AthleticWear 5.80 [5.31, 6.29] 1.06
##            DressedUp 3.75 [3.16, 4.34] 1.25
##          LeisureWear 2.90 [2.47, 3.33] 0.91
##
##   Application:Tinder
##               Outfit
##         AthleticWear 3.90 [3.42, 4.38] 1.02
##            DressedUp 5.80 [5.28, 6.32] 1.11
##          LeisureWear 3.05 [2.49, 3.61] 1.19
##
## Note. M and SD represent mean and standard deviation, respectively.
## LL and UL indicate the lower and upper limits of the
## 95% confidence interval for the mean, respectively.
## The confidence interval is a plausible range of population means
## that could have created a sample mean (Cumming, 2014).
```